

HTML5

Forms — Complete Reference

Forms Attributes Elements Input Types Validation

Everything you need to know about HTML forms — from the basics to advanced input types, validation, accessibility, and best practices.

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SECTION 1

The Element

Structure Purpose How Forms Work

What is an HTML Form?

An HTML form is a section of a document that contains interactive controls — text fields, checkboxes, radio buttons, dropdowns, and buttons — that allow users to submit data to a web server. Forms are the primary way users interact with web applications: login pages, registration, search bars, contact forms, checkout pages — all use HTML forms.

1.1 Basic Form Structure

```
<form action='/submit' method='POST'>

  <label for='username'>Username:</label>
  <input type='text' id='username' name='username'>

  <label for='password'>Password:</label>
  <input type='password' id='password' name='password'>

  <button type='submit'>Log In</button>

</form>
```

1.2 How Form Submission Works

When a user clicks Submit, the browser collects all input values and sends them to the URL specified in the action attribute using the method specified (GET or POST).

Step	What happens
1. User fills form	User types values into input fields
2. User clicks Submit	Browser intercepts the click event
3. Browser validates	HTML5 validation runs (required, pattern, min/max...)
4. Data is packaged	Each input's name=value is collected into form data
5. Request is sent	Browser sends HTTP GET or POST to action URL
6. Server processes	Server-side code (PHP, Node, Django...) handles data
7. Response returned	Server sends back HTML, JSON, or redirect

1.3 GET vs POST

Feature	GET	POST
Data location	Appended to URL: /search?q=html	Sent in request body (invisible)
Visibility	Visible in browser URL bar	Not visible in URL
Bookmark-able	Yes — URL contains all data	No
Data length	Limited (~2000 chars)	No practical limit
Security	Avoid for passwords/sensitive data	Safe for sensitive data
Use for	Search, filters, pagination	Login, registration, file upload, payments
Cached	Yes (browsers cache GET)	No
History	Stays in browser history	Not saved in history

```
<form action='/search' method='GET'>
  <input type='text' name='q' placeholder='Search...'>
  <button type='submit'>Search</button>
</form>

<form action='/login' method='POST'>
  <input type='email' name='email'>
  <input type='password' name='password'>
  <button type='submit'>Login</button>
</form>
```

SECTION 2

Form Attributes

action method enctype target novalidate autocomplete

All Attributes Explained

Attribute	Values	Purpose	Example
action	URL (relative or absolute)	Where to send form data on submit	action='/api/register'
method	GET POST	HTTP method used to send data	method='POST'
enctype	See below	How to encode data (important for files)	enctype='multipart/form-data'
target	_self _blank _parent _top	Where to display the response	target='_blank'
novalidate	(boolean)	Disable HTML5 built-in validation	novalidate
autocomplete	on off	Browser saves & suggests previous values	autocomplete='off'
name	text	Names the form for DOM/JS access	name='loginForm'
accept-charset	UTF-8 ISO-8859-1	Character encoding for submitted data	accept-charset='UTF-8'
rel	noopener noreferrer	Relationship of linked resource	rel='noopener'

2.1 enctype — Encoding Types

The enctype attribute determines how the form data is encoded before sending. It only applies when method='POST'. Getting this wrong will break file uploads.

enctype value	When to use	What it does
application/x-www-form-urlencoded	Default — all forms without files	Encodes spaces as + and special chars as %XX
multipart/form-data	REQUIRED for file uploads	Sends data in parts; preserves binary file data
text/plain	Debugging only — avoid in production	Sends data as plain text, no encoding

```
<form action='/contact' method='POST'
      enctype='application/x-www-form-urlencoded'>
  <input type='text' name='message'>
  <button type='submit'>Send</button>
</form>

<form action='/upload' method='POST'
      enctype='multipart/form-data'>
  <input type='file' name='avatar' accept='image/*'>
  <button type='submit'>Upload</button>
</form>

<form action='/results' method='GET' target='_blank'>
  <input type='text' name='q'>
  <button type='submit'>Search in new tab</button>
</form>

<form action='/submit' method='POST' novalidate>
  <input type='email' name='email' required>
  <button type='submit'>Submit</button>
</form>
```

TIP:
autocomplete

Use `autocomplete='off'` for OTP codes, CVV fields, and one-time fields. Leave it 'on' for name/address fields — it improves user experience.

SECTION 3

Form Elements

input textarea select button fieldset datalist output

All HTML Form Elements

Element	Purpose	Self-closing?	Key attributes
	Single-line data entry (many types)	Yes	type, name, value, placeholder
	Multi-line text entry	No	rows, cols, maxlength, wrap
	Dropdown list	No	multiple, size, name
	Item inside select or datalist	No	value, selected, disabled
	Groups options inside select	No	label, disabled
	Clickable button	No	type (submit/reset/button)
	Accessible text for an input	No	for (links to input id)
	Groups related form controls	No	disabled, name
	Caption/title for a fieldset	No	(content only)
	Predefined suggestions for input	No	id (linked to input list=)
	Displays result of a calculation	No	for, name, form
	Scalar measurement in known range	No	value, min, max, low, high
	Progress bar for a task	No	value, max

3.1 textarea

```
<label for='bio'>Biography:</label>
<textarea id='bio' name='bio'
  rows='5'
  cols='40'
```

```
        maxlength='500'  
        minlength='10'  
        placeholder='Tell us about yourself...'  
        wrap='soft'>  
    Default text goes here between tags  
</textarea>
```

3.2 select & option & optgroup

```
<label for='country'>Country:</label>  
<select id='country' name='country'>  
    <option value=''>-- Choose a country --</option>  
    <option value='us'>United States</option>  
    <option value='uk'>United Kingdom</option>  
    <option value='in' selected>India</option>  
    <option value='au' disabled>Australia (unavailable)</option>  
</select>  
  
<select name='car'>  
    <optgroup label='European'>  
        <option value='bmw'>BMW</option>  
        <option value='audi'>Audi</option>  
    </optgroup>  
    <optgroup label='Japanese'>  
        <option value='toyota'>Toyota</option>  
        <option value='honda'>Honda</option>  
    </optgroup>  
</select>  
  
<select name='skills' multiple size='4'>  
    <option value='html'>HTML</option>  
    <option value='css'>CSS</option>  
    <option value='js'>JavaScript</option>  
    <option value='py'>Python</option>  
</select>
```

3.3 button types

type value	Behaviour	When to use
submit	Submits the form data to action URL	Primary form submission button

reset	Clears all form fields to their default values	Use sparingly — users hate accidental resets
button	Does nothing by default — use with JS onclick	Triggering JavaScript actions

```
<form action='/submit' method='POST'>
  <input type='text' name='q'>

  <button type='submit'>Submit Form</button>

  <button type='reset'>Clear</button>

  <button type='button' onclick='previewForm()'>Preview</button>

  <input type='submit' value='Send'>
</form>
```

3.4 fieldset & legend

```
<form action='/register' method='POST'>

  <fieldset>
    <legend>Personal Information</legend>
    <label for='fname'>First Name:</label>
    <input type='text' id='fname' name='fname'>
    <label for='lname'>Last Name:</label>
    <input type='text' id='lname' name='lname'>
  </fieldset>

  <fieldset>
    <legend>Account Details</legend>
    <label for='email'>Email:</label>
    <input type='email' id='email' name='email'>
    <label for='pass'>Password:</label>
    <input type='password' id='pass' name='pass'>
  </fieldset>

  <fieldset disabled>
```

```
<legend>Billing (coming soon)</legend>

<input type='text' name='card'>

</fieldset>

<button type='submit'>Register</button>

</form>
```

3.5 datalist — Autocomplete Suggestions

```
<label for='browser'>Your browser:</label>
<input list='browsers' id='browser' name='browser'
      placeholder='Type or choose...'>

<datalist id='browsers'>  <!-- id must match input list= -->
  <option value='Chrome'>
  <option value='Firefox'>
  <option value='Safari'>
  <option value='Edge'>
  <option value='Opera'>
</datalist>
```

3.6 output, meter, progress

```
<form oninput='result.value = parseInt(a.value) + parseInt(b.value)'>
  <input type='number' id='a' name='a' value='0'>
  +
  <input type='number' id='b' name='b' value='0'>
  = <output name='result' for='a b'>0</output>
</form>

<label>Disk usage:</label>
<meter value='70' min='0' max='100' low='30' high='80' optimum='20'>
  70%
</meter>

<label>Uploading: <progress value='45' max='100'>45%</progress></label>

<progress>Loading...</progress>
```

SECTION 4

Input Types

All 22 HTML5 input types with examples and browser behaviour

Complete Input Type Reference

type	What it renders	Key behaviour	Browser UI
text	Single-line text field	Default type; accepts any text	Plain text box
password	Masked text field	Characters shown as dots/asterisks	Masked input
email	Text field for email address	Validates name@domain.tld format	@ keyboard on mobile
number	Numeric input	Allows min, max, step; rejects non-numbers	Spinner arrows
tel	Phone number field	No built-in validation (formats vary)	Dial-pad on mobile
url	URL input	Validates starts with http:// etc.	URL keyboard on mobile
search	Search field	Same as text but styled as search box	X clear button
date	Date picker	Returns YYYY-MM-DD; supports min/max	Calendar picker
time	Time picker	Returns HH:MM; 24h or 12h per locale	Time picker
datetime-local	Date + time picker (no timezone)	Returns YYYY-MM-DDTHH:MM	Combined picker
month	Month + year picker	Returns YYYY-MM	Month picker
week	Week number picker	Returns YYYY-Www	Week picker
color	Color picker	Returns hex value like #ff0000	Color swatch
range	Slider control	Returns number; use min/max/step	Drag slider
checkbox	Tick box	Checked=on; unchecked=not sent	Checkbox

radio	Single-select from group	Same name= groups them; one selectable	Radio button
file	File chooser	Use enctype multipart/form-data	Browse button
hidden	Invisible field	Sends data without showing user	Nothing (hidden)
submit	Submit button	Submits the form	Button
reset	Reset button	Clears all form fields	Button
button	Generic button	No default action; use with JS	Button
image	Image as submit button	Submits x,y coordinates of click	Image

4.1 Text & Password

```

<input type='text'
      id='username'
      name='username'
      placeholder='Enter username'
      value='JohnDoe'
      maxlength='20'
      size='30'>

<input type='password'
      id='pass'
      name='password'
      minlength='8'
      autocomplete='current-password'>

<input type='search'
      name='q'
      placeholder='Search products...'
      list='suggestions'>

```

4.2 Email, Tel, URL

```

<input type='email'
      name='email'
      placeholder='you@example.com'
      multiple>    <!-- multiple: allow comma-separated emails -->

```

```
<input type='tel'
      name='phone'
      placeholder='+91 98765 43210'
      pattern='[0-9]{10}'
      title='10-digit number'>

<input type='url'
      name='website'
      placeholder='https://example.com'>
```

4.3 Number & Range

```
<input type='number'
      name='quantity'
      min='1'
      max='100'
      step='1'
      value='1'>

<input type='number' name='price' min='0' max='9999' step='0.01'>

<input type='range'
      name='rating'
      min='0'
      max='10'
      step='1'
      value='5'
      oninput='ratingDisplay.value = this.value'>
<output name='ratingDisplay'>5</output>
```

4.4 Date & Time Types

```
<input type='date'
      name='birthday'
      min='1900-01-01'
      max='2010-12-31'>

<input type='time'
      name='meeting_time'
      min='09:00'
      max='18:00'>
```

```
        step='1800'> <!-- step in seconds: 1800 = 30 min increments -->

<input type='datetime-local'
        name='event_datetime'
        min='2025-01-01T00:00'
        max='2025-12-31T23:59'>

<input type='month' name='exp_month' min='2025-01'>

<input type='week' name='delivery_week'>
```

4.5 Checkbox & Radio

```
<fieldset>
  <legend>Skills (select all that apply):</legend>
  <input type='checkbox' id='html' name='skills' value='html' checked>
  <label for='html'>HTML</label>

  <input type='checkbox' id='css' name='skills' value='css'>
  <label for='css'>CSS</label>

  <input type='checkbox' id='js' name='skills' value='js'>
  <label for='js'>JavaScript</label>
</fieldset>

<fieldset>
  <legend>Gender:</legend>
  <input type='radio' id='male' name='gender' value='male'>
  <label for='male'>Male</label>

  <input type='radio' id='female' name='gender' value='female'>
  <label for='female'>Female</label>

  <input type='radio' id='other' name='gender' value='other' checked>
  <label for='other'>Other</label>
</fieldset>
```

4.6 File Upload

```
<form action='/upload' method='POST' enctype='multipart/form-data'>
```

```
<input type='file' name='resume' accept='.pdf,.doc,.docx'>

<input type='file' name='photo' accept='image/*'>

<input type='file' name='doc' accept='application/pdf,image/jpeg'>

<input type='file' name='gallery' accept='image/*' multiple>

<input type='file' name='selfie' accept='image/*' capture='user'>

<button type='submit'>Upload</button>

</form>
```

4.7 Color, Hidden, Image

```
<label for='theme'>Theme color:</label>
<input type='color' id='theme' name='theme_color' value='#3B82F6'>

<input type='hidden' name='csrf_token' value='xK9mP2qR8'>
<input type='hidden' name='user_id' value='12345'>

<input type='image'
  src='/images/submit-btn.png'
  alt='Submit'
  width='100'
  height='40'>
```

SECTION 5

Input Attributes

Every attribute for with values and examples

5.1 Universal Input Attributes

These attributes work on most or all input types.

Attribute	Values / Type	Description	Example
type	text email ...	Defines the input type (see Section 4)	type='email'
name	text	Key used when submitting data to server	name='username'
id	text	Unique identifier; links to label for=	id='email'
value	text/number	Pre-filled default value	value='John'
placeholder	text	Hint text shown when field is empty	placeholder='Enter name'
required	(boolean)	Field must be filled before submit	required
disabled	(boolean)	Field shown but not editable or sent	disabled
readonly	(boolean)	Value shown and SENT but not editable	readonly
autofocus	(boolean)	Field gets focus when page loads	autofocus
autocomplete	on off name ...	Controls browser autocomplete	autocomplete='email'
tabindex	number	Tab key order; -1 removes from order	tabindex='2'
form	form id	Associate input with a form by its id	form='myForm'
hidden	(boolean)	Hides element (still in DOM)	hidden

5.2 Text & String Attributes

Attribute	Value(s)	Description	Example
-----------	----------	-------------	---------

maxlength	number	Max characters allowed	maxlength='50'
minlength	number	Min characters required	minlength='8'
size	number	Visible width in characters	size='30'
pattern	regex	RegEx the value must match	pattern='[A-Z]{3}'
title	text	Tooltip shown on hover / error hint	title='3 uppercase letters'
spellcheck	true false	Enable/disable browser spellcheck	spellcheck='true'
inputmode	text numeric email . ..	Hint for mobile keyboard type	inputmode='numeric'
list	datalist id	Links to datalist for suggestions	list='cities'

5.3 Number & Range Attributes

Attribute	Value(s)	Description	Example
min	number/date	Minimum acceptable value	min='0'
max	number/date	Maximum acceptable value	max='100'
step	number any	Legal number intervals; 'any' = any value	step='0.5'

```
<input type='number' name='price' min='0' max='9999.99' step='0.01'>
```

```
<input type='number' name='age' min='18' max='120' step='1'>
```

```
<input type='range' name='rating' min='1' max='5' step='1' value='3'>
```

```
<input type='date' name='appt' min='2025-01-01' max='2025-12-31'>
```

5.4 File Attributes

Attribute	Value(s)	Description	Example
accept	MIME or ext	Filter allowed file types	accept='image/*,.pdf'
multiple	(boolean)	Allow multiple file selection	multiple
capture	user environment	Use camera (mobile); user=front, environment=back	capture='user'

5.5 Form Override Attributes (formXxx)

These attributes on a submit button override the parent form's settings for that specific submission. Extremely useful for multi-action forms.

Attribute	Overrides	Example
formaction	form action=	formaction='/save-draft'
formmethod	form method=	formmethod='GET'
formenctype	form enctype=	formenctype='multipart/form-data'
formtarget	form target=	formtarget='_blank'
formnovalidate	form novalidate	formnovalidate

```
<form action='/submit' method='POST'>
  <input type='text' name='title'>
  <textarea name='content'></textarea>

  <button type='submit'>Publish</button>

  <button type='submit'
    formaction='/draft'
    formmethod='POST'>Save Draft</button>

  <button type='submit'
    formaction='/preview'
    formnovalidate
    formtarget='_blank'>Preview</button>
</form>
```

SECTION 6

HTML5 Form Validation

Built-in validation pattern custom messages CSS states

6.1 Built-in Validation Attributes

Attribute	Works on	What it checks	Error message trigger
required	All inputs	Field is not empty	Please fill out this field
minlength	text, password, etc.	Min number of characters	Please use at least N characters
maxlength	text, password, etc.	Max characters (silently limits)	(prevents typing past limit)
min	number, date, range	Value >= min	Value must be >= N
max	number, date, range	Value <= max	Value must be <= N
pattern	text, email, url, tel	Value matches regex	Please match the format: [title]
type=email	email	Contains @ and valid domain	Please enter a valid email
type=url	url	Starts with valid protocol	Please enter a valid URL
type=number	number	Value is a number	Please enter a valid value
step	number, range	Value is a valid step increment	Please enter a valid value

6.2 Pattern Attribute — Common Regex Examples

Use case	pattern value	What it matches
Uppercase only (3 chars)	[A-Z]{3}	Exactly 3 uppercase letters
Alphanumeric username	[a-zA-Z0-9_]{3,20}	3-20 letters, digits, underscores
Indian phone number	[6-9][0-9]{9}	10 digits starting with 6-9
Strong password	(?=.*\d)(?=.*[a-z])(?=.*[A-Z]).{8,}	8+ chars, upper, lower, digit
Pincode (India)	[1-9][0-9]{5}	6-digit Indian postal code
Date (DD/MM/YYYY)	(0[1-9] [12][0-9] 3[01])/(0[1-9] 1[012])/[0-9]{4}	Day/Month/Year format

Hex color code	<code>#[0-9A-Fa-f]{6}</code>	<code>#RRGGBB</code> format
IP address (basic)	<code>(\d{1,3}\.){3}\d{1,3}</code>	Basic IPv4 format

```

<input type='email' name='email' required
      placeholder='you@example.com'>

<input type='password' name='pass'
      required
      minlength='8'
      pattern='(?=.*[a-z])(?=.*[A-Z])(?=.*[0-9]).{8,}'
      title='At least 8 chars with uppercase, lowercase, and number'>

<input type='tel' name='mobile'
      pattern='[6-9][0-9]{9}'
      title='Enter valid 10-digit Indian mobile number'
      placeholder='9876543210'>

<input type='number' name='rating'
      min='1' max='5' step='0.5' required>

```

6.3 CSS Validation States (Pseudo-classes)

CSS Pseudo-class	When it applies	Typical styling
<code>:valid</code>	Input value passes validation	Green border / green checkmark
<code>:invalid</code>	Input value fails validation	Red border (be careful — shows on page load)
<code>:required</code>	Input has required attribute	Asterisk marker or bold label
<code>:optional</code>	Input does NOT have required	Normal styling
<code>:focus</code>	Input is currently focused / active	Blue outline / highlight
<code>:disabled</code>	Input has disabled attribute	Grey out, cursor not-allowed
<code>:read-only</code>	Input has readonly attribute	Slightly greyed, no cursor
<code>:checked</code>	Checkbox/radio is checked	Custom tick mark styling
<code>:in-range</code>	Number/range value within min-max	Green indicator
<code>:out-of-range</code>	Number/range value outside min-max	Red indicator

:placeholder-shown	Placeholder text is visible (empty)	Style empty-state label effect
:user-valid	Valid after user has interacted (new)	Safer than :valid — no flash on load
:user-invalid	Invalid after user has interacted	Show error only after typing

```

/* CSS for live validation feedback */

/* Only show errors AFTER user has interacted (modern browsers) */
input:user-invalid {
  border: 2px solid #EF4444;
  background: #FEF2F2;
}

input:user-valid {
  border: 2px solid #10B981;
  background: #ECFDF5;
}

/* Fallback for older browsers */
input:invalid:not(:placeholder-shown) {
  border: 2px solid #EF4444;
}

/* Style required fields */
input:required { border-left: 3px solid #F97316; }

/* Disabled field */
input:disabled {
  background: #F3F4F6;
  cursor: not-allowed;
  opacity: 0.6;
}

```

6.4 Custom Validation Messages via JavaScript

```

<input type='email' id='email' name='email' required>
<span id='emailError' class='error'></span>

<script>
  const emailInput = document.getElementById('email');

```

```
emailInput.addEventListener('invalid', function(e) {
    e.preventDefault(); // stop browser default popup
    if (this.validity.valueMissing) {
        document.getElementById('emailError').textContent
            = 'Email address is required.';
    } else if (this.validity.typeMismatch) {
        document.getElementById('emailError').textContent
            = 'Please enter a valid email like: you@example.com';
    }
});

emailInput.addEventListener('input', function() {
    document.getElementById('emailError').textContent = '';
    this.setCustomValidity(''); // clear custom error
});
</script>
```

SECTION 7

Best Practices

Accessibility Security UX Semantics Full Example

7.1 Accessibility (a11y)

Accessible forms work with screen readers, keyboard navigation, and assistive technology. This is both a legal requirement in many countries and good practice.

Rule	Why	Example
Always use	Screen readers announce label when input focused	Name:
Never use placeholder as label	Placeholder disappears on type — confuses users	Use label + placeholder together
Use fieldset + legend	Groups related fields semantically for SR	Address
Add aria-describedby	Links hint text / error message to input	aria-describedby='emailHint'
Add aria-required='true'	Extra signal for screen readers	aria-required='true'
Visible focus styles	Keyboard users need to see where they are	:focus { outline: 2px solid blue }
Error messages near input	Screen reader users miss distant errors	span role=alert — place error near field
Logical tab order	Tab key should flow top-to-bottom, left-to-right	Avoid tabindex > 0 where possible

7.2 Security Best Practices

Practice	Why it matters	Implementation
Always validate server-side	Client HTML can be bypassed entirely	Never trust only HTML5 validation
Use HTTPS	HTTP sends form data in plain text	SSL/TLS certificate required
CSRF tokens	Prevents cross-site request forgery attacks	
Sanitise all input	Prevents XSS (script injection) attacks	Server: escape HTML entities
Use POST for sensitive data	GET exposes data in URL and server logs	Login/register/payment = POST

Limit file types & size	Prevents malicious file upload attacks	Server-side check, not just accept=
Autocomplete='off' on OTP	Stops browser from storing one-time codes	autocomplete='one-time-code'

7.3 Complete Real-World Registration Form

```
<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>
  <title>Registration</title>
</head>
<body>

<form id='registerForm'
      action='/api/register'
      method='POST'
      enctype='application/x-www-form-urlencoded'
      autocomplete='on'
      novalidate>

  <input type='hidden' name='csrf_token' value='abc123xyz'>

  <fieldset>
    <legend>Personal Information</legend>

    <label for='fname'>First Name <span aria-hidden='true'>*</span></label>
    <input type='text'
          id='fname' name='first_name'
          required minlength='2' maxlength='50'
          autocomplete='given-name'
          aria-required='true'
          placeholder='Alice'>

    <label for='lname'>Last Name <span aria-hidden='true'>*</span></label>
    <input type='text'
          id='lname' name='last_name'
          required minlength='2' maxlength='50'
          autocomplete='family-name'
```

```
placeholder='Smith'>

<label for='dob'>Date of Birth</label>
<input type='date'
      id='dob' name='dob'
      min='1900-01-01'
      max='2007-12-31'>

<label>Gender</label>
<input type='radio' id='g-male' name='gender' value='male'>
<label for='g-male'>Male</label>
<input type='radio' id='g-female' name='gender' value='female'>
<label for='g-female'>Female</label>
<input type='radio' id='g-other' name='gender' value='other'>
<label for='g-other'>Other</label>
</fieldset>

<fieldset>
  <legend>Account Details</legend>

  <label for='email'>Email Address <span aria-hidden='true'>*</span></label>
  <input type='email'
        id='email' name='email'
        required
        autocomplete='email'
        aria-describedby='emailHint'
        placeholder='you@example.com'>
  <small id='emailHint'>We will never share your email.</small>

  <label for='mobile'>Mobile Number</label>
  <input type='tel'
        id='mobile' name='mobile'
        pattern='[6-9][0-9]{9}'
        title='Enter 10-digit Indian mobile number'
        autocomplete='tel'
        placeholder='9876543210'>

  <label for='pass'>Password <span aria-hidden='true'>*</span></label>
  <input type='password'
        id='pass' name='password'
        required minlength='8'
```

```
        pattern='(?=.*[a-z])(?=.*[A-Z])(?=.*[0-9]).{8,}'
        title='Min 8 chars: uppercase, lowercase, and digit'
        autocomplete='new-password'
        aria-describedby='passHint'>
    <small id='passHint'>Min 8 characters with uppercase and digit.</small>
</fieldset>

<fieldset>
    <legend>Preferences</legend>

    <label for='country'>Country</label>
    <select id='country' name='country' autocomplete='country-name'>
        <option value=''>Select country...</option>
        <option value='IN'>India</option>
        <option value='US'>United States</option>
        <option value='UK'>United Kingdom</option>
    </select>

    <label for='bio'>Short Bio</label>
    <textarea id='bio' name='bio'
        rows='3' maxlength='300'
        placeholder='Tell us about yourself (optional)'>
    </textarea>

    <input type='checkbox' id='terms' name='terms' value='agreed' required>
    <label for='terms'>I agree to the Terms and Conditions *</label>

    <input type='checkbox' id='newsletter' name='newsletter' value='yes'>
    <label for='newsletter'>Subscribe to newsletter</label>
</fieldset>

<button type='submit'>Create Account</button>
<button type='reset'>Clear Form</button>

</form>
</body>
</html>
```

Quick Reference Cheat Sheet

Form Element Attributes

action='/url' method='GET|POST'
 enctype='multipart/form-data' (files)
 target='_blank' novalidate
 autocomplete='on|off' name='id'

All Input Types

text password email number tel url
 search date time datetime-local
 month week color range
 checkbox radio file hidden
 submit reset button image

Validation Attributes

required minlength='n'
 maxlength='n' min='n' max='n'
 step='n' pattern='regex'
 title='hint' type=email|url|number

Common Input Attributes

name='key' value='default'
 id='uid' placeholder='hint'
 disabled readonly autofocus
 tabindex='n' form='formId'
 list='datalistId' autocomplete='...'

Form Elements

input textarea select button
 label fieldset legend datalist
 option optgroup output meter progress

CSS Validation States

:valid :invalid :required
 :disabled :read-only :checked
 :focus :placeholder-shown
 :in-range :out-of-range
 :user-valid :user-invalid (modern)

Common Mistakes to Avoid

Using == on Strings in HTML	Nothing — HTML is markup, not code
Missing label for= / input id=	Screen readers cannot identify the field
Using placeholder as label	Placeholder disappears on type — use
GET method for login/passwords	Password appears in URL and server logs
Forgetting enctype for file upload	File data is lost / corrupted
Relying only on client validation	JS can be disabled; always validate server-side
Missing name= attribute	That field's value is NOT sent to the server
Using BR tags for form layout	Use CSS Grid / Flexbox for form layout